Robert's Feedback as Formula Proposals

This note translates Robert's comments into mathematical variants that can be folded into the AD-CISPO / CW-CISPO proposal.

Glossary

Symbol	Meaning	Notes
q	Prompt or question	Fixed input before generation starts.
i	Response index	One sampled response within a group of completions for the same prompt.
G	Group size	Number of sampled responses for one prompt.
o_i	Sampled response i	Generated token sequence.
$o_{i,t}$	Token at position t in response i	A generated token, not a prompt token, when $t \in M_i$.
T_i	Length of response i	Number of generated-token positions before masking.
t	Target token position	Position whose saliency, credit weight, or clip multiplier is being computed.
j	Future token position	Used when a later token attends back to position t.
u	Dummy summation index	Used when averaging over all valid positions in M_i.
$s_{i,t}$	Autoregressive state before token $o_{i,t}$	Prompt plus generated prefix before position t.
M_i	Valid generated-token action positions for response i	Excludes prompt tokens, padding, masked-out actions, special tokens, and sink tokens removed by the saliency guard. All token-level means, sums, and mean-one normalizations are over this set.
π_θ	Current policy being optimized	The model receiving the policy-gradient update.
$\pi_{\theta_{\text{old}}}$	Rollout policy	The policy that produced the sampled batch.
$\rho_{i,t}$	Token importance-sampling ratio	Current-policy probability divided by rollout-policy probability for token $o_{i,t}$.
$\hat{A}_i$	Sequence-level advantage	Usually group-relative for the sampled response.

Symbol	Meaning	Notes
r_i	Outcome reward for response i	Usually a non-differentiable verifier reward.
F	Teacher-forced scalar objective	Current causal-tangent proxy differentiates this objective, not reward directly.
$h_{i,t}^{(\ell)}$	Hidden state	Layer ℓ, response i, position t.
$S_{i,t}$	Generic token saliency score	Sections below define attention, causal-tangent, rollout, entropy, and opportunity variants.
$H_{i,t}$	Token entropy	Uncertainty in the next-token distribution.
$C_{i,t}$	Controllability score	How much action choice at a state changes downstream return.
$w_{i,t}$	CW-CISPO token-credit weight	Detached, bounded, and normalized to mean one over M_i.
$m_{i,t}$	AD-CISPO clip multiplier	Used to redistribute the upper clipping radius.
P_i	Token attention transition matrix	Used for PageRank-style multi-step attention saliency.
c_i	PageRank-style token centrality vector	Fixed-point centrality over generated-token positions.

Shared Notation

For prompt q, let response index i select one sampled completion:

$$o_i = (o_{i,1}, \ldots, o_{i,T_i}). \tag{1}$$

Let

$$s_{i,t} = (q, o_{i,<t}) \tag{2}$$

be the autoregressive state before token $o_{i,t}$.

The set M_i is the set of valid generated-token action positions for response i. It is the denominator set for token-level means and the support for credit weights. In practice, M_i excludes prompt positions, padding positions, masked-out generated actions, special tokens, and separator or punctuation sink tokens removed by the saliency guard. Thus M_i is the set of generated positions eligible to receive saliency mass, CW-CISPO credit weight, or AD-CISPO clipping budget.

The token importance-sampling ratio is

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid s_{i,t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid s_{i,t})}. \tag{3}$$

1. Current Proxy: Teacher-Forced Objective Sensitivity

The current causal-tangent saliency is approximately a hidden-state sensitivity to a teacher-forced scalar objective F, not directly to the outcome reward.

One clean way to write the current objective is

$$F_i(\theta) = |\hat{A}_i| \frac{1}{|M_i|} \sum_{t \in M_i} \log \pi_\theta(o_{i,t} \mid s_{i,t}), \tag{4}$$

with batch objective

$$F(\theta) = \sum_i F_i(\theta). \tag{5}$$

The current causal-tangent score is then

$$S_{i,t}^F = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \frac{\left| \nabla_{h_{i,t}^{(\ell)}} F \cdot \left(h_{i,t}^{(\ell)} - \frac{1}{|M_i|} \sum_{u \in M_i} h_{i,u}^{(\ell)} \right) \right|}{\sqrt{d_{\text{model}}}}. \tag{6}$$

In words:

$$S_{i,t}^F \approx \text{local effect of changing } h_{i,t} \text{ on teacher-forced objective } F. \tag{7}$$

Robert's point is that the desired object is closer to reward sensitivity:

$$S_{i,t}^r \approx \left\| \nabla_{h_{i,t}} r_i \right\|. \tag{8}$$

Since r_i is usually a non-differentiable verifier reward, the differentiable ideal is more like

$$S_{i,t}^r \approx \left\| \nabla_{h_{i,t}} \mathbb{E}_{o_{i,>t} \sim \pi_\theta} \left[r(q, o_{i,\le t}, o_{i,>t}) \right] \right\|. \tag{9}$$

2. Rollout-Based Reward Impact, VinePPO-Style

The expensive but conceptually clean target is the counterfactual effect of token $o_{i,t}$ on final reward. Define the continuation value after prefix $o_{i,\le t}$:

$$Q^\pi(s_{i,t}, o_{i,t}) = \mathbb{E}_{o_{i,>t} \sim \pi} \left[r(q, o_{i,\le t}, o_{i,>t}) \right]. \tag{10}$$

Define the pre-token value

$$V^\pi(s_{i,t}) = \mathbb{E}_{a \sim \pi(\cdot | s_{i,t})} \left[Q^\pi(s_{i,t}, a) \right]. \tag{11}$$

Then a reward-impact saliency is

$$S_{i,t}^{\text{rollout}} = \left| Q^\pi(s_{i,t}, o_{i,t}) - V^\pi(s_{i,t}) \right|. \tag{12}$$

With K sampled continuations per candidate token, estimate

$$\widehat{Q}^\pi(s_{i,t}, a) = \frac{1}{K} \sum_{k=1}^{K} r(q, o_{i,<t}, a, o_{>t}^{(k)}). \tag{13}$$

The rollout saliency estimator becomes

$$\widehat{S}_{i,t}^{\text{rollout}} = \left| \widehat{Q}^\pi(s_{i,t}, o_{i,t}) - \sum_a \pi(a \mid s_{i,t}) \widehat{Q}^\pi(s_{i,t}, a) \right|. \tag{14}$$

3. Learned Offline Predictor for Grad-H F or Reward Impact

Instead of computing expensive saliency online every batch, train a small predictor g_ϕ from hidden states and cheap token features.

For vector-gradient prediction:

$$g_\phi(h_{i,t}^{(\ell)}, x_{i,t}) \approx \mathrm{sg}\left(\nabla_{h_{i,t}^{(\ell)}} F\right), \tag{15}$$

with loss

$$\mathcal{L}_{\mathrm{grad}}(\phi) = \mathbb{E}_{i,t,\ell}\left[\left\| g_\phi(h_{i,t}^{(\ell)}, x_{i,t}) - \mathrm{sg}\left(\nabla_{h_{i,t}^{(\ell)}} F\right)\right\|_2^2\right]. \tag{16}$$

For scalar saliency prediction:

$$\hat{S}_{i,t}^{\phi} = g_\phi(h_{i,t}, x_{i,t}) \approx S_{i,t}^{F} \quad \text{or} \quad S_{i,t}^{r}. \tag{17}$$

Train with

$$\mathcal{L}_{\mathrm{score}}(\phi) = \mathbb{E}_{i,t}\left[\left(g_\phi(h_{i,t}, x_{i,t}) - \mathrm{sg}(S_{i,t}^{\star})\right)^2\right], \tag{18}$$

where

$$S_{i,t}^{\star} \in \left\{S_{i,t}^{F}, S_{i,t}^{r}, S_{i,t}^{\mathrm{rollout}}\right\}. \tag{19}$$

Then CW-CISPO can use

$$w_{i,t} = \mathrm{ProjectMeanOne}\left(\left(\hat{S}_{i,t}^{\phi} + \epsilon\right)^{\gamma}\right). \tag{20}$$

4. Probability-Space Sensitivity

Robert was unsure about saliency written as $\nabla_{p(o_i)} F$. The concern is that perturbing the probability of a sampled token is not the same as perturbing a hidden state or a policy parameter.

Let

$$p_{i,t} = \pi_\theta(o_{i,t} \mid s_{i,t}). \tag{21}$$

A probability-coordinate sensitivity would be

$$S_{i,t}^{p} = \left| p_{i,t} \frac{\partial F}{\partial p_{i,t}} \right|. \tag{22}$$

Equivalently, because

$$\frac{\partial F}{\partial \log p_{i,t}} = p_{i,t} \frac{\partial F}{\partial p_{i,t}}, \tag{23}$$

we can write

$$S_{i,t}^{p} = \left| \frac{\partial F}{\partial \log \pi_\theta(o_{i,t} \mid s_{i,t})} \right|. \tag{24}$$

For the simple teacher-forced objective above,

$$\frac{\partial F}{\partial \log \pi_\theta(o_{i,t} \mid s_{i,t})} = \frac{|\hat{A}_i|}{|M_i|}. \tag{25}$$

That is nearly uniform within a response, so it is not a useful token saliency unless F includes richer token-level structure.

5. Decision-State and Bottleneck-State Formulations

In deep RL terms, a token position is important if the next action materially changes the reachable return distribution.

A decision-state score is

$$D_{i,t} = \mathrm{Var}_{a\sim\pi(\cdot|s_{i,t})}\left[Q^{\pi}(s_{i,t}, a)\right]. \tag{26}$$

Equivalently, a max-gap version is

$$D_{i,t}^{\max} = \max_{a} Q^{\pi}(s_{i,t}, a) - \min_{a} Q^{\pi}(s_{i,t}, a). \tag{27}$$

A bottleneck-state score can be written as the association between visiting a state-like feature z and eventual success:

$$B(z) = \Pr(z \in \tau \mid r = 1) - \Pr(z \in \tau \mid r = 0). \tag{28}$$

For token positions, define a feature map $\psi(s_{i,t})$. Then

$$B_{i,t} = B(\psi(s_{i,t})). \tag{29}$$

The doorway analogy from grid-world RL corresponds to states with both high visitation centrality and high success association:

$$S_{i,t}^{\text{bottleneck}} = C_{i,t}^{\text{visit}} \cdot |B_{i,t}|. \tag{30}$$

6. Multi-Step Attention Centrality, PageRank-Style

Current future-attention in-degree is a one-step score:

$$S_{i,t}^{\text{attn},1} = \sum_{\ell,h} \sum_{\substack{j\in M_i \\ j>t}} a_{i,j,t}^{(\ell,h)}. \tag{31}$$

This implicitly weights every future query token j equally.

Define a row-stochastic token transition matrix P_i over generated positions:

$$P_i[j,t] = \frac{1}{|L_{\text{top}}|} \sum_{\ell\in L_{\text{top}}} \frac{1}{H_\ell} \sum_{h=1}^{H_\ell} a_{i,j,t}^{(\ell,h)}. \tag{32}$$

A PageRank-style equilibrium centrality c_i satisfies

$$c_i = (1-\alpha)\nu_i + \alpha P_i^{\top} c_i, \qquad 0 < \alpha < 1, \tag{33}$$

where ν_i can be uniform over M_i:

$$\nu_i[t] = \frac{1}{|M_i|}. \tag{34}$$

The multi-step attention saliency is

$$S_{i,t}^{\text{attn,PR}} = c_i[t]. \tag{35}$$

Equivalently,

$$c_i = (1-\alpha) \sum_{n=0}^{\infty} \alpha^n \left(P_i^{\top}\right)^n \nu_i. \tag{36}$$

This is attention paid by future tokens, weighted by those tokens' own centrality.

7. Multiplicative Saliency, Entropy, and Controllability

The current saliency-only form is

$$\tilde{w}_{i,t} = (S_{i,t} + \epsilon)^{\gamma}. \tag{37}$$

Robert's suggestion is closer to a multiplicative gate: update tokens only when they are both important and changeable.

Let token entropy be

$$H_{i,t} = -\sum_{a} \pi_\theta(a \mid s_{i,t}) \log \pi_\theta(a \mid s_{i,t}). \tag{38}$$

Let controllability be

$$C_{i,t} = \mathrm{Var}_{a \sim \pi_\theta(\cdot \mid s_{i,t})} \left[Q^{\pi}(s_{i,t}, a) \right]. \tag{39}$$

A simple multiplicative weight is

$$\tilde{w}^{\mathrm{mult}}_{i,t} = (S_{i,t} + \epsilon)^{\gamma} \, (H_{i,t} + \epsilon)^{\eta} \, (C_{i,t} + \epsilon)^{\kappa}. \tag{40}$$

Then enforce bounded mean-one credit weights:

$$w^{\mathrm{mult}}_{i,t} = \mathrm{ProjectMeanOne}\left(\tilde{w}^{\mathrm{mult}}_{i,t}; w_{\min}, w_{\max} \right), \tag{41}$$

so that

$$\frac{1}{|M_i|} \sum_{t \in M_i} w^{\mathrm{mult}}_{i,t} = 1. \tag{42}$$

An Osband-style delight gate can be written as a sigmoid:

$$g_{i,t} = \sigma\left(\beta_0 + \beta_\ell \left[-\log \pi_\theta(o_{i,t} \mid s_{i,t}) \right] + \beta_A \hat{A}_i + \beta_C \log(C_{i,t} + \epsilon) \right). \tag{43}$$

Then

$$\tilde{w}^{\mathrm{gate}}_{i,t} = (S_{i,t} + \epsilon)^{\gamma} \, g_{i,t}. \tag{44}$$

The intuition is

$$\text{credit} \approx \text{importance} \times \text{uncertainty} \times \text{controllability}. \tag{45}$$

8. Cumulative Future Entropy Baseline

A cheap baseline is cumulative future entropy: tokens are important if many uncertain choices remain downstream.

Define entropy-to-go:

$$E^{\mathrm{future}}_{i,t} = \sum_{\substack{j \in M_i \\ j \geq t}} \lambda^{j-t} H_{i,j}, \qquad 0 < \lambda \leq 1. \tag{46}$$

The corresponding mean-one credit weight is

$$w^{\mathrm{futureH}}_{i,t} = \mathrm{ProjectMeanOne}\left(\left(E^{\mathrm{future}}_{i,t} + \epsilon \right)^{\eta} \right). \tag{47}$$

9. Token-Dependent Learning Rate View

CW-CISPO weights can also be read as a token-dependent learning rate.

The scalar CISPO gradient has the form

$$g_{\mathrm{CISPO}} = -\sum_{i,t} \mathrm{sg}(\hat{\rho}_{i,t}) \hat{A}_i \nabla_\theta \log \pi_\theta(o_{i,t} \mid s_{i,t}). \tag{48}$$

CW-CISPO changes it to

$$g_{\mathrm{CW}} = -\sum_{i,t} \mathrm{sg}(w_{i,t}) \, \mathrm{sg}(\hat{\rho}_{i,t}) \hat{A}_i \nabla_\theta \log \pi_\theta(o_{i,t} \mid s_{i,t}). \tag{49}$$

This is equivalent to a per-token step size

$$\eta_{i,t} = \eta_0 w_{i,t}. \tag{50}$$

Mean-one normalization preserves the average learning rate:

$$\frac{1}{|M_i|} \sum_{t \in M_i} \eta_{i,t} = \eta_0. \tag{51}$$

Robert's learning-rate point can be stated as an implicit relative-step decay:

$$\frac{\|\Delta\theta\|_2}{\|\theta\|_2} \approx \eta \frac{\|g\|_2}{\|\theta\|_2}. \tag{52}$$

Even with constant η, this relative step shrinks as $\|\theta\|_2$ grows. Token-dependent weights can selectively counteract or amplify this decay at important states.

10. Feature-Space Opportunity Proxy

Robert's "opportunities opened by this token" idea can be formalized as successor-feature dispersion.

Let $\phi(q, o_{\le t}, o_{>t})$ be a feature vector of the completed trajectory, such as final hidden state, answer embedding, or verifier-side features. Define the feature opportunity of prefix $o_{\le t}$:

$$O_{i,t} = \mathrm{tr}\left(\mathrm{Cov}_{o_{>t} \sim \pi}\left[\phi(q, o_{\le t}, o_{>t})\right]\right). \tag{53}$$

An action-conditional opportunity gain is

$$\Delta O_{i,t} = O(q, o_{\le t}) - \mathbb{E}_{a \sim \pi(\cdot \mid s_{i,t})}\left[O(q, o_{<t}, a)\right]. \tag{54}$$

Use

$$S^{\mathrm{opp}}_{i,t} = |\Delta O_{i,t}| \tag{55}$$

as a proxy for whether token $o_{i,t}$ opens or closes useful future trajectory options.

11. Golden Trajectory, Teacher Logits, and PRM Access

There are three increasingly rich supervision signals.

Golden trajectory only. Given only a teacher or gold trajectory o^T, the objective is

$$F_{\text{gold}}(\theta) = \sum_t \log \pi_\theta(o_t^T \mid q, o_{<t}^T). \tag{56}$$

The saliency is

$$S_t^{\text{gold}} = \|\nabla_{h_t} F_{\text{gold}}\|. \tag{57}$$

Teacher logits. If teacher logits or a teacher distribution $\tau_t(a)$ are available, use

$$F_{\text{logits}}(\theta) = \sum_t \sum_a \tau_t(a) \log \pi_\theta(a \mid q, o_{<t}^T). \tag{58}$$

Equivalently,

$$F_{\text{logits}}(\theta) = -\sum_t \text{KL}\left(\tau_t(\cdot) \,\|\, \pi_\theta(\cdot \mid q, o_{<t}^T)\right) + \text{const}. \tag{59}$$

The saliency is

$$S_t^{\text{logits}} = \|\nabla_{h_t} F_{\text{logits}}\|. \tag{60}$$

Process reward model. If a process reward model R_ψ is available, define

$$F_{\text{PRM}}(\theta) = \sum_t R_\psi(q, o_{\le t}). \tag{61}$$

The saliency is

$$S_t^{\text{PRM}} = \|\nabla_{h_t} F_{\text{PRM}}\|. \tag{62}$$

The hierarchy is therefore

$$\text{gold trajectory} \;<\; \text{teacher logits} \;\not\equiv\; \text{process reward model}. \tag{63}$$

12. Cost of Full-Trajectory Grad-H F

The current causal-tangent formulation requires a full teacher-forced forward pass over the sampled trajectory:

$$(q, o_i) \longmapsto \{h_{i,t}^{(\ell)}\}_{\ell \in L_{\text{top}},\, t \in M_i} \longmapsto F_i \longmapsto \nabla_{h_{i,t}^{(\ell)}} F_i. \tag{64}$$

The cost scales as

$$\text{cost}\left(S^F\right) \approx \text{Forward}(T_i) + \text{Backward}_h(T_i, |L_{\text{top}}|), \tag{65}$$

with activation storage

$$\text{memory}\left(S^F\right) = O\left(|L_{\text{top}}|\, T_i\, d_{\text{model}}\right). \tag{66}$$

This is why cheap baselines are important:

$$S_{i,t}^{\text{attn},1}, \qquad S_{i,t}^{\text{attn,PR}}, \qquad S_{i,t}^{\text{futureH}}, \qquad \hat{S}_{i,t}^{\phi}. \tag{67}$$

13. Baseline Matrix

The concrete baseline family can be summarized as

$$S_{i,t}^{\mathrm{attn},1} = \sum_{\ell,h} \sum_{j>t} a_{i,j,t}^{(\ell,h)}, \tag{68}$$

$$S_{i,t}^{\mathrm{attn,PR}} = c_i[t], \quad c_i = (1-\alpha)\nu_i + \alpha P_i^{\top} c_i, \tag{69}$$

$$S_{i,t}^{F} = \left\| \nabla_{h_{i,t}} F \right\|_{\mathrm{tangent}}, \tag{70}$$

$$S_{i,t}^{r} = \left\| \nabla_{h_{i,t}} \mathbb{E}[r \mid q, o_{\le t}] \right\|, \tag{71}$$

$$S_{i,t}^{\mathrm{rollout}} = \left| Q^{\pi}(s_{i,t}, o_{i,t}) - V^{\pi}(s_{i,t}) \right|, \tag{72}$$

$$S_{i,t}^{\mathrm{futureH}} = \sum_{j \ge t} \lambda^{j-t} H_{i,j}, \tag{73}$$

$$S_{i,t}^{\mathrm{opp}} = \left| \Delta O_{i,t} \right|. \tag{74}$$

Each score becomes a CISPO credit weight through the same bounded mean-one map:

$$w_{i,t} = \mathrm{ProjectMeanOne}\left((S_{i,t} + \epsilon)^{\gamma}; w_{\min}, w_{\max} \right). \tag{75}$$

The main experimental question is:

$$\text{Does } w(S^{F}),\ w(S^{r}),\ \text{or } w(S^{\mathrm{rollout}}) \text{ beat } w(S^{\mathrm{futureH}}) \text{ and shuffled controls?} \tag{76}$$